

ISyE 6525 HW3 — Question 2

Q2(a) Upper bound on tensor rank

Let

$$\mathcal{X} \in \mathbb{R}^{I \times J \times K}.$$

The CP rank of $\mathcal{X}$ is the minimum number of rank-one tensors whose sum equals $\mathcal{X}$.

We first show

$$\text{rank}(\mathcal{X}) \leq IJ.$$

Let

$$e_i^{(I)} \in \mathbb{R}^I, \quad e_j^{(J)} \in \mathbb{R}^J$$

be standard basis vectors. For example,

$$e_2^{(4)} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}.$$

Also let

$$x_{ij:} = \mathcal{X}_{ij:} \in \mathbb{R}^K$$

be the mode-3 fiber obtained by fixing indices i and j .

For a fixed pair (i, j) , consider

$$e_i^{(I)} \circ e_j^{(J)} \circ x_{ij:}.$$

This tensor is zero everywhere except along the fiber at location $(i, j, :)$, where it reproduces the original fiber $x_{ij:}$.

Entrywise,

$$\left(e_i^{(I)} \circ e_j^{(J)} \circ x_{ij:} \right)_{abc} = (e_i^{(I)})_a (e_j^{(J)})_b X_{ijc}.$$

Since the standard basis vectors are one-hot,

$$(e_i^{(I)})_a (e_j^{(J)})_b = \begin{cases} 1, & a = i, b = j, \\ 0, & \text{otherwise.} \end{cases}$$

Therefore,

$$\mathcal{X} = \sum_{i=1}^I \sum_{j=1}^J e_i^{(I)} \circ e_j^{(J)} \circ x_{ij:}.$$

Each nonzero summand is an outer product of three vectors and therefore has rank one. There are at most IJ such terms, so

$$\boxed{\text{rank}(\mathcal{X}) \leq IJ}.$$

The same argument can be repeated using the other two families of fibers.

Using mode-1 fibers $x_{:jk} \in \mathbb{R}^I$,

$$\mathcal{X} = \sum_{j=1}^J \sum_{k=1}^K x_{:jk} \circ e_j^{(J)} \circ e_k^{(K)},$$

so

$$\text{rank}(\mathcal{X}) \leq JK.$$

Using mode-2 fibers $x_{i:k} \in \mathbb{R}^J$,

$$\mathcal{X} = \sum_{i=1}^I \sum_{k=1}^K e_i^{(I)} \circ x_{i:k} \circ e_k^{(K)},$$

so

$$\text{rank}(\mathcal{X}) \leq IK.$$

Since all three bounds hold simultaneously,

$$\boxed{\text{rank}(\mathcal{X}) \leq \min\{IJ, JK, IK\}}.$$

Q2(b) CP decomposition as a special case of Tucker

Suppose $\mathcal{X}$ has the CP representation

$$\mathcal{X} = \sum_{r=1}^R \lambda_r a_r \circ b_r \circ c_r.$$

Define the factor matrices

$$A = \begin{bmatrix} a_1 & a_2 & \cdots & a_R \end{bmatrix} \in \mathbb{R}^{I \times R},$$

$$B = \begin{bmatrix} b_1 & b_2 & \cdots & b_R \end{bmatrix} \in \mathbb{R}^{J \times R},$$

and

$$C = \begin{bmatrix} c_1 & c_2 & \cdots & c_R \end{bmatrix} \in \mathbb{R}^{K \times R}.$$

Now define a Tucker core

$$\mathcal{G} \in \mathbb{R}^{R \times R \times R}$$

that is zero everywhere except on its superdiagonal:

$$g_{pqr} = \begin{cases} \lambda_r, & p = q = r, \\ 0, & \text{otherwise.} \end{cases}$$

A Tucker model expands as

$$\mathcal{G} \times_1 A \times_2 B \times_3 C = \sum_{p=1}^R \sum_{q=1}^R \sum_{r=1}^R g_{pqr} a_p \circ b_q \circ c_r.$$

Because $g_{pqr} = 0$ unless $p = q = r$, all cross-component terms vanish, leaving only

$$\mathcal{G} \times_1 A \times_2 B \times_3 C = \sum_{r=1}^R \lambda_r a_r \circ b_r \circ c_r = \mathcal{X}.$$

Hence

CP is a special case of Tucker with a superdiagonal core.

The conceptual difference is that a general Tucker core can contain entries such as

$$g_{123} \neq 0,$$

which produces an interaction term

$$a_1 \circ b_2 \circ c_3.$$

CP forbids such cross-component interactions because its core is restricted to the superdiagonal.

Parameter counts

For a fair comparison, take Tucker multilinear rank (R, R, R) .

Using the raw scalar-coefficient count used by the course for factor/core storage:

CP

The factor matrices contain

$$IR + JR + KR$$

entries, and the weights contribute another R . Therefore,

$$N_{\text{CP}} = R(I + J + K + 1).$$

General Tucker

The factor matrices again contain

$$IR + JR + KR$$

entries, while a general $R \times R \times R$ core contains

$$R^3$$

entries. Therefore,

$$N_{\text{Tucker}} = R(I + J + K) + R^3.$$

For the special Tucker representation corresponding to CP, only the R superdiagonal core entries are nonzero/free, so its coefficient count reduces to the CP count.

Note on the meaning of “free parameters”

The counts above are the raw model coefficients, which is the convention used here for comparing the two representations.

If the CP factor columns are explicitly constrained to unit norm, then each vector loses one continuous degree of freedom. Under that stricter independent-degree-of-freedom interpretation,

$$N_{\text{CP,DOF}} = R[(I - 1) + (J - 1) + (K - 1) + 1] = R(I + J + K - 2).$$

The remaining sign and component-permutation ambiguities are discrete and do not change the continuous dimension.